

Exercise-2

Problem statement: Implementing Ecommerce platform search function

Source code:

```
package ds;

import java.util.Arrays;
import java.util.Comparator;

public class EcommerceSearch {

    public static class Product {

        private int productId;

        private String productName;

        private String category;

        public Product(int productId, String productName, String category) {

            this.productId = productId;

            this.productName = productName;

            this.category = category;

        }

        public int getProductId() {

            return productId;

        }

        public String getProductName() {

            return productName;

        }

        public String getCategory() {

            return category;

        }

    }

}
```

```

@Override

public String toString() {

    return "Product{ID=" + productId + ", Name=" + productName + ", Category=" +
category + "}";

}

}

public static Product linearSearch(Product[] products, int targetId) {

    for (Product product : products) {

        if (product.getProductId() == targetId) {

            return product;

        }

    }

    return null;

}

public static Product binarySearch(Product[] sortedProducts, int targetId) {

    int left = 0;

    int right = sortedProducts.length - 1;

    while (left <= right) {

        int mid = left + (right - left) / 2;

        if (sortedProducts[mid].getProductId() == targetId) {

            return sortedProducts[mid];

        } else if (sortedProducts[mid].getProductId() < targetId) {

            left = mid + 1;

        } else {

            right=mid-1;

        }

    }

}

```

```

    }
    return null;
}

public static void main(String[] args) {
    Product[] inventory = {
        new Product(105, "Wireless Mouse", "Electronics"),
        new Product(101, "Mechanical Keyboard", "Electronics"),
        new Product(108, "Running Shoes", "Apparel"),
        new Product(103, "Coffee Mug", "Kitchen"),
        new Product(102, "Desk Lamp", "Home Office")
    };

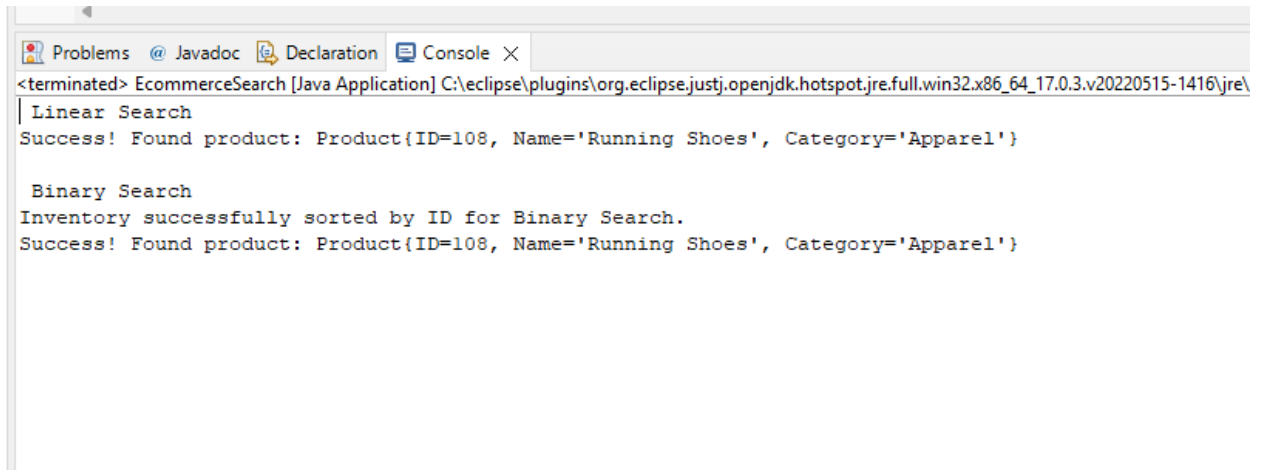
    int targetId = 108;
    System.out.println(" Linear Search");
    Product linearResult = linearSearch(inventory, targetId);
    printResult(targetId, linearResult);
    System.out.println("\n Binary Search");
    Arrays.sort(inventory, Comparator.comparingInt(Product::getProductId));
    System.out.println("Inventory successfully sorted by ID for Binary Search.");
    Product binaryResult = binarySearch(inventory, targetId);
    printResult(targetId, binaryResult);
}

private static void printResult(int targetId, Product result) {
    if (result != null) {
        System.out.println("Success! Found product: " + result);
    } else {
        System.out.println("Product with ID " + targetId + " not found.");
    }
}

```

```
}  
}
```

Output:



```
<terminated> EcommerceSearch [Java Application] C:\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.3.v20220515-1416\jre\
| Linear Search
Success! Found product: Product{ID=108, Name='Running Shoes', Category='Apparel'}

Binary Search
Inventory successfully sorted by ID for Binary Search.
Success! Found product: Product{ID=108, Name='Running Shoes', Category='Apparel'}
```